

CHAPTER 4 – EXPLORING ALGEBRAIC IDENTITIES

EXERCISE SET 4.3

IMPORTANT IDENTITIES

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

1. Find the following squares using one of the above identities. Determine which identity makes the calculation easier.

(i) 117^2

(ii) 78^2

(iii) 198^2

(iv) 214^2

(v) 1104^2

(vi) 1120^2

(i)

$$117^2$$

Write

$$117 = 100 + 17.$$

Using

$$(a + b)^2 = a^2 + 2ab + b^2,$$

we get

$$\begin{aligned} 117^2 &= (100 + 17)^2 \\ &= 100^2 + 2(100)(17) + 17^2 \\ &= 10000 + 3400 + 289 \end{aligned}$$

$$\boxed{117^2 = 13689}$$

(ii)

$$78^2$$

Write

$$78 = 80 - 2.$$

Using

$$(a - b)^2 = a^2 - 2ab + b^2,$$

$$\begin{aligned} 78^2 &= (80 - 2)^2 \\ &= 80^2 - 2(80)(2) + 2^2 \\ &= 6400 - 320 + 4 \end{aligned}$$

$$\boxed{78^2 = 6084}$$

(iii)

$$198^2$$

Write

$$198 = 200 - 2.$$

Therefore,

$$\begin{aligned} 198^2 &= (200 - 2)^2 \\ &= 200^2 - 2(200)(2) + 2^2 \\ &= 40000 - 800 + 4 \end{aligned}$$

$$\boxed{198^2 = 39204}$$

(iv)

$$214^2$$

Write

$$214 = 200 + 14.$$

Using

$$\begin{aligned} (a + b)^2 &= a^2 + 2ab + b^2, \\ 214^2 &= (200 + 14)^2 \\ &= 200^2 + 2(200)(14) + 14^2 \\ &= 40000 + 5600 + 196 \end{aligned}$$

$$\boxed{214^2 = 45796}$$

(v)

$$1104^2$$

Write

$$1104 = 1100 + 4.$$

Therefore,

$$\begin{aligned} 1104^2 &= (1100 + 4)^2 \\ &= 1100^2 + 2(1100)(4) + 4^2 \\ &= 1210000 + 8800 + 16 \end{aligned}$$

$$\boxed{1104^2 = 1218816}$$

(vi)

$$1120^2$$

Write

$$1120 = 1100 + 20.$$

Therefore,

$$\begin{aligned} 1120^2 &= (1100 + 20)^2 \\ &= 1100^2 + 2(1100)(20) + 20^2 \\ &= 1210000 + 44000 + 400 \end{aligned}$$

$$\boxed{1120^2 = 1254400}$$

2. Factor using suitable identities:

(i) $16y^2 - 24y + 9$

(ii) $\frac{9}{4}s^2 + 6st + 4t^2$

(iii) $\frac{m^2}{9} + \frac{mk}{3} + \frac{k^2}{4} + 3nk + 2mn + 9n^2$

(iv) $\frac{p^2}{16} - 2 + \frac{16}{p^2}$

(v) $9a^2 + 4b^2 + c^2 - 12ab + 6ac - 4bc$

(i)

$$16y^2 - 24y + 9$$

We have

$$16y^2 = (4y)^2, \quad 9 = 3^2$$

and

$$24y = 2(4y)(3).$$

Therefore,

$$16y^2 - 24y + 9$$

$$= (4y)^2 - 2(4y)(3) + 3^2$$

Using

$$a^2 - 2ab + b^2 = (a - b)^2,$$

we get

$$\boxed{(4y - 3)^2}.$$

(ii)

$$\frac{9}{4}s^2 + 6st + 4t^2$$

We have

$$\frac{9}{4}s^2 = \left(\frac{3}{2}s\right)^2$$

and

$$4t^2 = (2t)^2.$$

Also,

$$2\left(\frac{3}{2}s\right)(2t) = 6st.$$

Therefore,

$$\begin{aligned} & \frac{9}{4}s^2 + 6st + 4t^2 \\ &= \left(\frac{3}{2}s\right)^2 + 2\left(\frac{3}{2}s\right)(2t) + (2t)^2 \\ &= \boxed{\left(\frac{3}{2}s + 2t\right)^2}. \end{aligned}$$

(iii)

$$\frac{m^2}{9} + \frac{mk}{3} + \frac{k^2}{4} + 3nk + 2mn + 9n^2$$

Group the terms as

$$\frac{m^2}{9} + \frac{k^2}{4} + 9n^2 + \frac{mk}{3} + 3nk + 2mn.$$

Now,

$$\frac{m^2}{9} = \left(\frac{m}{3}\right)^2,$$

$$\frac{k^2}{4} = \left(\frac{k}{2}\right)^2,$$

and

$$9n^2 = (3n)^2.$$

Also,

$$2\left(\frac{m}{3}\right)\left(\frac{k}{2}\right) = \frac{mk}{3},$$

$$2\left(\frac{k}{2}\right)(3n) = 3nk,$$

and

$$2(3n)\left(\frac{m}{3}\right) = 2mn.$$

Therefore,

$$\begin{aligned} & \frac{m^2}{9} + \frac{mk}{3} + \frac{k^2}{4} + 3nk + 2mn + 9n^2 \\ &= \left(\frac{m}{3}\right)^2 + \left(\frac{k}{2}\right)^2 + (3n)^2 \\ & \quad + 2\left(\frac{m}{3}\right)\left(\frac{k}{2}\right) + 2\left(\frac{k}{2}\right)(3n) + 2(3n)\left(\frac{m}{3}\right). \end{aligned}$$

Using

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca,$$

we get

$$\boxed{\left(\frac{m}{3} + \frac{k}{2} + 3n\right)^2}.$$

(iv)

$$\frac{p^2}{16} - 2 + \frac{16}{p^2}$$

Observe that

$$\frac{p^2}{16} = \left(\frac{p}{4}\right)^2$$

and

$$\frac{16}{p^2} = \left(\frac{4}{p}\right)^2.$$

Also,

$$2 \left(\frac{p}{4}\right) \left(\frac{4}{p}\right) = 2.$$

Therefore,

$$\begin{aligned} & \frac{p^2}{16} - 2 + \frac{16}{p^2} \\ &= \left(\frac{p}{4}\right)^2 - 2 \left(\frac{p}{4}\right) \left(\frac{4}{p}\right) + \left(\frac{4}{p}\right)^2. \end{aligned}$$

Using

$$(a - b)^2 = a^2 - 2ab + b^2,$$

we get

$$\boxed{\left(\frac{p}{4} - \frac{4}{p}\right)^2}.$$

(v)

$$9a^2 + 4b^2 + c^2 - 12ab + 6ac - 4bc$$

Observe that

$$9a^2 = (3a)^2,$$

$$4b^2 = (-2b)^2,$$

and

$$c^2 = c^2.$$

Now,

$$2(3a)(-2b) = -12ab,$$

$$2(3a)(c) = 6ac,$$

and

$$2(-2b)(c) = -4bc.$$

Therefore,

$$\begin{aligned} &9a^2 + 4b^2 + c^2 - 12ab + 6ac - 4bc \\ &= (3a)^2 + (-2b)^2 + c^2 \\ &\quad + 2(3a)(-2b) + 2(3a)(c) + 2(-2b)(c). \end{aligned}$$

Using

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca,$$

we get

$$\boxed{(3a - 2b + c)^2}.$$

3. Expand the following using the identity

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca.$$

(i) $(p + 3q + 7r)^2$

(ii) $(3x - 2y + 4z)^2$

(i)

$$(p + 3q + 7r)^2$$

Here,

$$a = p, \quad b = 3q, \quad c = 7r.$$

Using

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca,$$

we get

$$\begin{aligned}(p + 3q + 7r)^2 &= p^2 + (3q)^2 + (7r)^2 \\ &\quad + 2(p)(3q) + 2(3q)(7r) + 2(7r)(p).\end{aligned}$$

Therefore,

$$\boxed{p^2 + 9q^2 + 49r^2 + 6pq + 42qr + 14pr}$$

(ii)

$$(3x - 2y + 4z)^2$$

Write

$$3x - 2y + 4z = 3x + (-2y) + 4z.$$

Thus,

$$a = 3x, \quad b = -2y, \quad c = 4z.$$

Using

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca,$$

we get

$$\begin{aligned}(3x - 2y + 4z)^2 &= (3x)^2 + (-2y)^2 + (4z)^2 \\ &\quad + 2(3x)(-2y) + 2(-2y)(4z) + 2(4z)(3x).\end{aligned}$$

Therefore,

$$\boxed{9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24xz}$$

4. Is the following an identity?

$$\begin{aligned}(a + b - c)^2 + (a - b + c)^2 + (a - b - c)^2 \\ = 2a^2 + 2b^2 + 2c^2.\end{aligned}$$

Solution

We expand each term separately.

First,

$$\begin{aligned}(a + b - c)^2 \\ = a^2 + b^2 + c^2 + 2ab - 2ac - 2bc.\end{aligned}$$

Second,

$$\begin{aligned}(a - b + c)^2 \\ = a^2 + b^2 + c^2 - 2ab + 2ac - 2bc.\end{aligned}$$

Third,

$$\begin{aligned}(a - b - c)^2 \\ = a^2 + b^2 + c^2 - 2ab - 2ac + 2bc.\end{aligned}$$

Adding the three expressions,

$$\begin{aligned}LHS &= 3a^2 + 3b^2 + 3c^2 \\ &\quad - 2ab - 2ac - 2bc.\end{aligned}$$

But the given right-hand side is

$$RHS = 2a^2 + 2b^2 + 2c^2.$$

Clearly,

$$LHS \neq RHS$$

for all values of a, b, c .

Therefore, the given statement is

NOT an identity.

QUICK REVISION – EXERCISE 4.3

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

Factorisation:

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = (a + b + c)^2$$